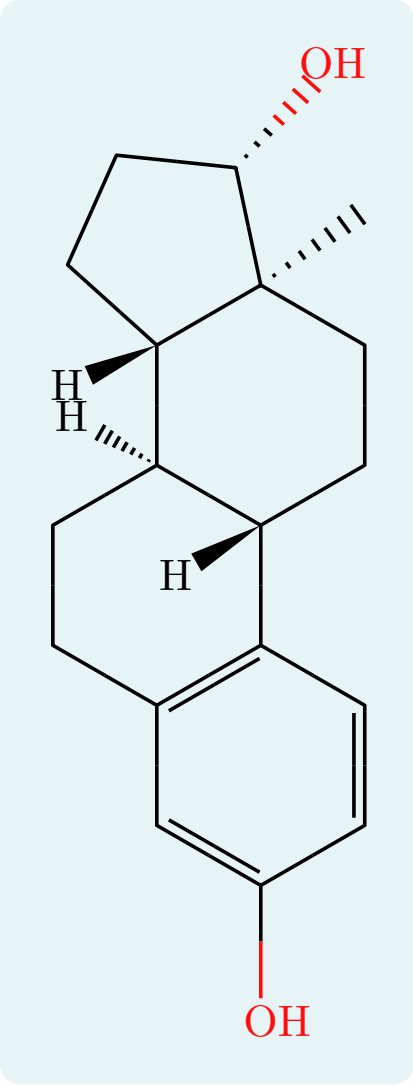
typed-smiles

Render SMILES strings as 2D molecular diagrams

User Guide



Version 0.7.0

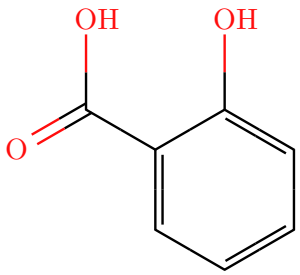
Contents

1 Introduction

typed-smiles renders SMILES strings as 2D skeletal molecular diagrams inside Typst documents. A bundled WebAssembly plugin parses the SMILES, computes atom coordinates, implicit hydrogens, and stereochemistry, and the Typst layer draws the bonds, atom labels, colors, charges, hydrogens, wedges, and reaction helpers.

This guide documents every function, argument, and package-specific syntax extension. Each feature comes with a runnable example: the source is on the left, its rendered output on the right.

```typst
#smiles("OC1=CC=CC=C1C(=O)O")
```



2 Getting started

2.1 Installation and import

Import the package from the Typst preview namespace. A wildcard import gives you every public symbol:

```typst
#import "@preview/typed-smiles:0.7.0": *
```

Typst

Or import only what you need:

```typst
#import "@preview/typed-smiles:0.7.0": smiles, ce, mol, rxn-arrow, reaction
```

Typst

The package exports these main symbols:

Symbol	Purpose
`smiles`	Render a SMILES string as a molecule (the main function).
`ce`	Chemical formulas and equations, re-exported from `chemformula`.
`mol`	Wrap a molecule with an optional caption.
`rxn-arrow`	A reaction arrow with conditions above and below.
`reaction`	Lay out a multi-step reaction scheme.
`atom`, `bond`, `lp`, `species`	References for arrows and highlights.
`arrow`, `highlight`	Mechanism arrows and shaded atom/bond regions.
`brackets`	Draw square brackets with optional corner marks.
`mol-weight`	Compute molecular weight from a SMILES string.

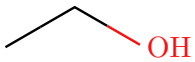
2.2 Your first molecule

Call `smiles` with a SMILES string.

```typst
Ethanol: #smiles("CCO")
```

Typst

Ethanol:



3 The `smiles()` function

`smiles` is the main function. Its full signature is:

```typst
#smiles(
  smiles-str,
  style: "default",
  scale: 1.0,
  bond-length: none,
  font-size: none,
  font: auto,
  bond-stroke: none,
  color: auto,
  fg: auto,
  theme: auto,
  rotation: 0deg,
  mirror: none,
  show-h: (),
  aromatic: "kekule",
  atom-annotations: (),
  opacity: 100%,
  bond-customizations: (),
  lone-pairs: none,
  atom-colors: (:),
  show-indices: false,
)
```

3.1 Argument reference

Argument	Type	Default	Description
`smiles-str`	`str`	(required)	The SMILES string.
`style`	`str`	`"default"`	Journal style preset (`"acs"`, `"rsc"`, `"nature"`, `"wiley"`) filling in bond length, label size, stroke, and font. Explicit arguments win; `"default"` applies nothing.
`scale`	`float`	`1.0`	Balanced scale for bond length, label size, and stroke at once.
`bond-length`	`float` / `none`	`none`	Bond length factor (`1.0` is 30 pt). Overrides `scale` for length.
`font-size`	`length` / `none`	`none`	Atom-label font size. Overrides `scale` for labels.

`font`	`auto` / `str` / `array`	`auto`	Font family for atom labels; `auto` is "New Computer Modern", or the preset's font when `style` is set.
`bond-stroke`	`length` / `none`	`none`	Bond stroke width. Overrides `scale` for strokes.
`color`	`auto` / `bool`	`auto`	Apply Jmol CPK atom colors. `auto` is colored for `"default"` and monochrome for journal presets.
`fg`	`auto` / `color`	`auto`	Foreground for bonds and carbon labels; `auto` inherits the surrounding text color.
`theme`	`auto` / `"light"` / `"dark"`	`auto`	CPK palette variant; `auto` picks "dark" when `fg` is light.
`rotation`	`angle`	`0deg`	Rotate the molecule; labels stay upright.
`mirror`	`none` / `"horizontal"` / `"vertical"`	`none`	Optional horizontal or vertical page-axis reflection.
`show-h`	`"all"` / `int` / `array`	`()`	Implicit hydrogens to label beyond the default heteroatom hydrogens. Use `"all"` for every atom.
`aromatic`	`"kekule"` / `"circle"`	`"kekule"`	Depict lowercase-aromatic rings with alternating double bonds or with an inscribed circle.
`atom-annotations`	`array`	`()`	Small gray side labels as `(index, content)` or `(index, content, offset)` tuple entries.
`opacity`	`ratio` / `float`	`100%`	Fade the whole drawing — bonds, labels, charges, lone pairs — e.g. for ghost molecules.
`bond-customizations`	`array`	`()`	Per-bond style overrides as `(bond(i, j), (..options..))` pairs; options are `color`, `stroke`, and `opacity`.

`lone-pairs`	`none` / `"dots"` / `"lines"`	`none`	Draw optional non-bonding electron pairs on skeletal atom labels.
`atom-colors`	`dictionary`	`(:)`	Color overrides for elements and labels. Element-symbol keys (e.g. `O: red`) override CPK colors; brace-quoted label keys (e.g. `"{PPh3}": purple`) override a specific abbreviated group. See Section 7.

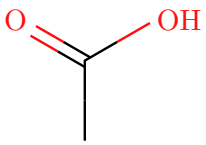
Note. `scale` sizes everything together. Use `bond-length`, `font-size`, and `bond-stroke` to override one dimension on its own; each defaults to `none`, meaning it follows `scale`.

3.2 `smiles-str`

The positional argument is the SMILES string.

```
1  #smiles("CC(=O)O")
```

Typst

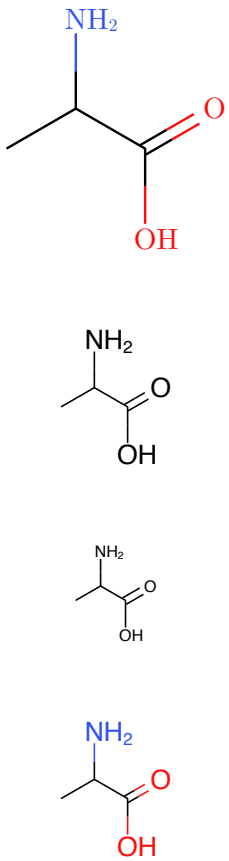


3.3 `style`

Journal style presets approximate the corresponding ChemDraw document stylesheets using the journals' published drawing settings — bond length, atom-label size, line width, a Helvetica/Arial font stack, and monochrome atom labels by default:

Preset	Bond	Line	Labels
`"acs"`	14.4 pt	0.6 pt	10 pt
`"rsc"`	12.2 pt	0.5 pt	7 pt
`"nature"`	10.8 pt	0.6 pt	6 pt
`"wiley"`	17 pt	0.7 pt	12 pt

```typst
#smiles("CC(N)C(=O)O") \
#smiles("CC(N)C(=O)O", style: "acs") \
#smiles("CC(N)C(=O)O", style: "nature") \
#smiles("CC(N)C(=O)O", style: "acs",
color: true)
```



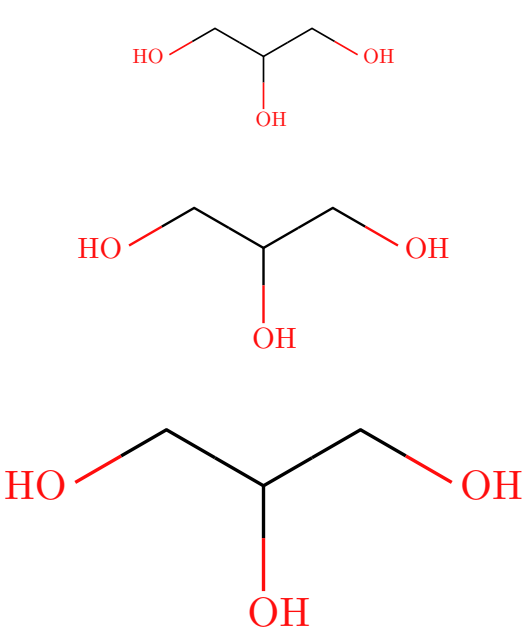
A preset only fills in arguments left unset: `style: "acs"` with `font-size: 14pt` keeps the ACS bond length under your label size, and `scale` multiplies the whole preset. `color: auto` makes journal presets monochrome; pass `color: true` for CPK colors or `color: false` for explicit black line art. `"default"` applies nothing, so existing documents are unaffected. Wiley's line width and label size are derived from its 6 mm bond-length guideline, which fixes only a minimum line width; journal-specific geometry beyond these four settings (double-bond spacing, hash spacing) keeps typed-smiles' defaults.

3.4 scale

Scales bond length, labels, and stroke proportionally.

```
#smiles("OCC(O)CO", scale: 0.7) \
#smiles("OCC(O)CO", scale: 1.0) \
#smiles("OCC(O)CO", scale: 1.4)
```

Typst

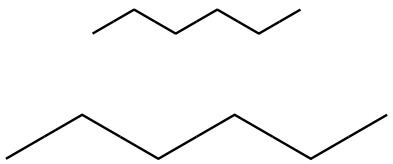


3.5 bond-length

Sets bond length on its own (`1.0` is 30 pt per bond), leaving label size and stroke under `scale`.

```
#smiles("CCCCCC", bond-length: 0.6) \
#smiles("CCCCCC", bond-length: 1.1)
```

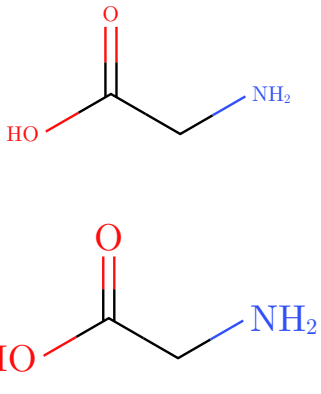
Typst



3.6 font-size

```
#smiles("OC(=O)CN", font-size: 8pt) \
#smiles("OC(=O)CN", font-size: 14pt)
```

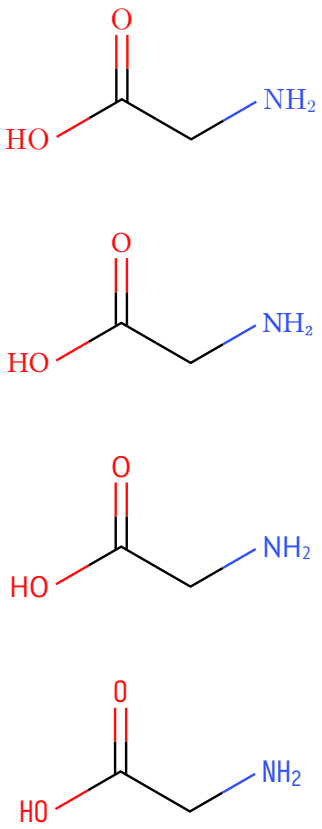
Typst



3.7 font

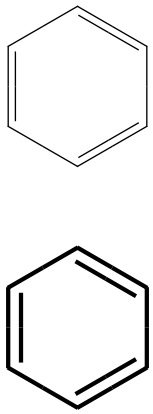
Match the document typeface, or pick something with character.

```typst
#smiles("OC(=O)CN", font: "New Computer Modern") \
#smiles("OC(=O)CN", font: "Libertinus Serif") \
#smiles("OC(=O)CN", font: "PT Sans") \
#smiles("OC(=O)CN", font: "Iosevka")
```



3.8 bond-stroke

```typst
#smiles("C1=CC=CC=C1", bond-stroke: 0.5pt) \
#smiles("C1=CC=CC=C1", bond-stroke: 1.6pt)
```

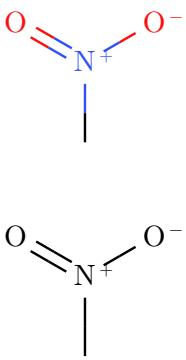


3.9 color

CPK colors are on by default for `style: "default"` (see Section 7). Journal presets default to monochrome line art. Set `color: false` for an explicit monochrome diagram, or `color: true` to force CPK colors.

```
#smiles("C[N+](=O)[O-]") \
#smiles("C[N+](=O)[O-]", color: false)
```

Typst

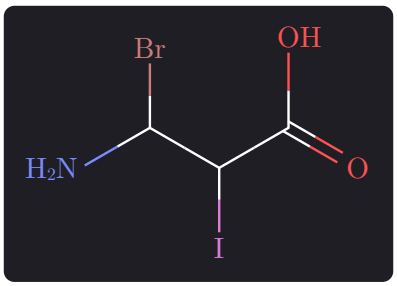


3.10 fg and theme

Bond strokes and carbon labels follow `fg`, which defaults to `auto` and inherits the surrounding text color — on a dark slide theme (Touying, Polylux, or any `set text(fill: white)`) molecules recolor with no configuration. `theme: auto` switches to a dark CPK variant whenever the foreground is light, lifting only the hues that vanish on dark backgrounds (N, O, Br, I, and dark named label colors such as navy or maroon).

```
#block(fill: rgb("#1E1E24"),
inset: 8pt,
  radius: 4pt)[
  #set text(fill: white)
  #smiles("NC(Br)C(I)C(=O)O")
]
```

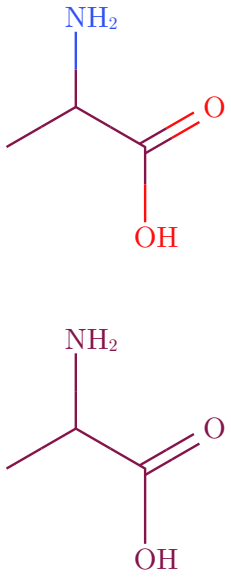
Typst



Both can be forced: an explicit `fg` recolors the skeleton on any background, and `theme: "dark"` or `theme: "light"` pins the palette. To set a document-wide default, rebind once in the preamble: `#let smiles = smiles.with(fg: white, theme: "dark")`.

```
#smiles("CC(N)C(=O)O", fg: maroon) \
#smiles("CC(N)C(=O)O", color: false, fg: maroon)
```

Typst



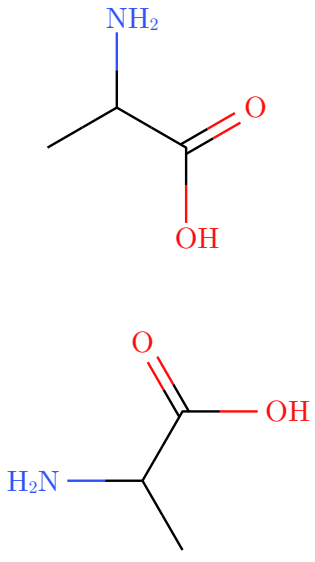
> **Note.** Reaction arrows inherit the surrounding text color the same way via `rxn-arrow(color: auto)`, so full schemes work on dark slides too.

3.11 `rotation`

Rotates the whole molecule while keeping every label upright.

```
#smiles("CC(N)C(=O)O", rotation: 0deg) \
#smiles("CC(N)C(=O)O", rotation: 90deg)
```

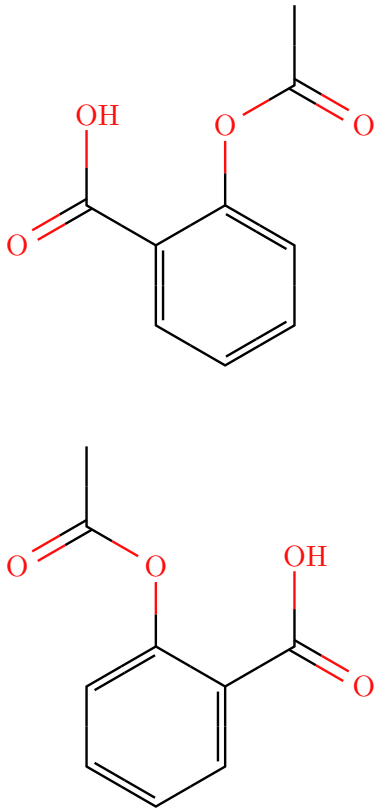
Typst



3.12 `mirror`

`mirror` reflects a molecule horizontally or vertically in the final drawing. Useful to face a reacting group toward a reaction arrow. Inside `reaction()`, pass it per molecule: `mol("OC1CCCCC1", mirror: "horizontal")`. With `rotation` set, `mirror: "vertical"` preserves left/right and flips top/bottom.

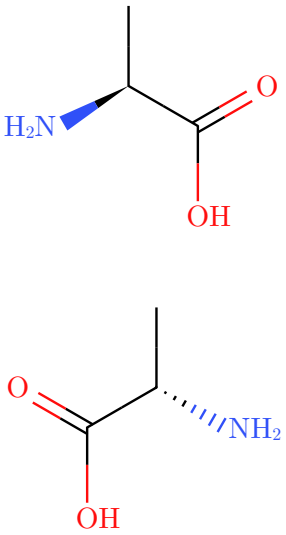
```
#smiles("CC(=O)OC1=CC=CC=C1C(=O)O") Typst
\
#smiles(
  "CC(=O)OC1=CC=CC=C1C(=O)O",
  mirror: "horizontal",
)
```



Note. Single-axis reflection exchanges wedges and hashes: a reflected 2D skeleton with unchanged wedges would depict the enantiomer, while reflecting and swapping is equivalent to turning the molecule over — the depicted configuration at every stereocenter is preserved.

A stereocenter keeps its configuration when mirrored.

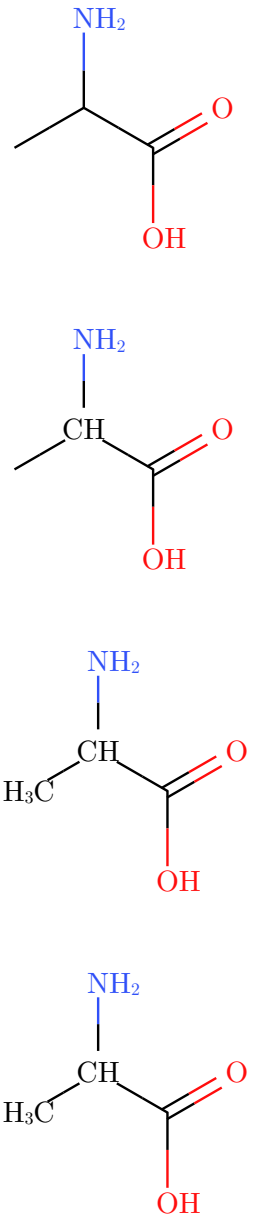
```typst
#smiles("N[C@@H](C)C(=O)O") \
#smiles(
  "N[C@@H](C)C(=O)O",
  mirror: "horizontal",
)
```



3.13 show-h

Hydrogens on heteroatoms are shown by default; carbon hydrogens are hidden. `show-h` selects carbon hydrogens without needing a trailing comma for one atom: use an integer for one atom, an array for several atoms, or `"all"` for every implicit hydrogen. Use `show-indices: true` to read off the indices while writing.

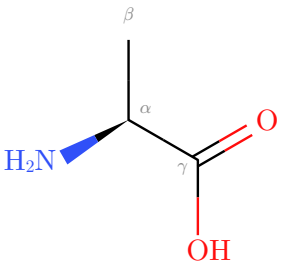
```typst
#smiles("CC(N)C(=O)O") \
#smiles("CC(N)C(=O)O", show-h: 1) \
#smiles("CC(N)C(=O)O", show-h: (0, 1)) \
#smiles("CC(N)C(=O)O", show-h: "all")
```



3.14 atom-annotations

Small annotations for NMR numbering, Greek positions, or footnote marks are drawn small and gray on the emptiest side of an atom. `atom-annotations` is always a tuple list. Each entry is either `(index, content)` or `(index, content, offset)`. Values are content, so wrap them in `text()` to restyle.

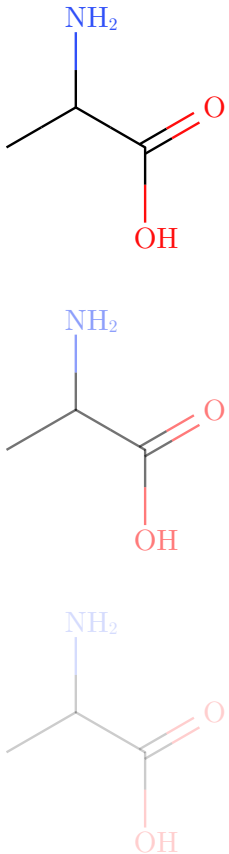
```typst
#smiles(
  "N[C@@H](C)C(=O)O",
  atom-annotations: (
    (1, [$alpha$], (0, -0.05)),
    (2, [$beta$]),
    (3, [$gamma$], (-0.4, -0.3)),
  ),
)
```



3.15 opacity

`opacity` fades every paint the drawing produces — bonds, atom labels, charges, lone pairs, and annotations — so a molecule can be ghosted or de-emphasized, e.g. spectator species in a mechanism or step-by-step slide builds. It accepts a ratio (`40%`) or a number between 0 and 1.

```typst
#smiles("CC(N)C(=O)O")
#h(1.5em)
#smiles("CC(N)C(=O)O", opacity: 55%)
#h(1.5em)
#smiles("CC(N)C(=O)O", opacity: 0.2)
```



Note. String molecules inside `reaction()` accept the same option: `mol("CCO", opacity: 30%)`.

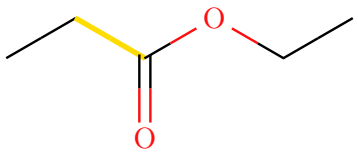
3.16 bond-customizations

`bond-customizations` restyles individual bonds. Each entry pairs a `bond(i, j)` reference (the same atom-index form used by `arrow()` and `highlight()` a plain `(i, j)` array also works) with an options dictionary:

- `color` — draw the whole bond in one color instead of the bicolor halves,
- `stroke` — the bond width, like a per-bond `bond-stroke`,
- `opacity` — fade just this bond.

Overrides apply to every part of the bond: both lines of a double bond, the hash lines of a stereo bond, waves and dashes.

```
#smiles(
  "CCC(=O)OCC",
  bond-customizations: (
    (bond(1, 2), (color: yellow, stroke: 2pt)),
  ),
)
```

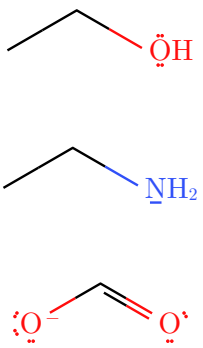


Note. Use `show-indices: true` while writing the references, exactly as for mechanism arrows. Bond geometry (length and angles) is computed by the layout engine and is not a per-bond style; use `bond-length` to size all bonds together.

3.17 lone-pairs

Set `lone-pairs` to `"dots"` or `"lines"` to annotate the skeletal drawing with non-bonding electron pairs on common organic heteroatoms and charged atoms. The default `none` keeps lone pairs hidden.

```
#smiles("CCO", lone-pairs: "dots") \
#smiles("CCN", lone-pairs: "lines") \
#smiles("[O-]C=O", lone-pairs: "dots")
```



Note. Lone pairs are inferred for a conservative organic subset such as N, O, S, P, halogens, and simple charged forms. They are an annotation on the skeletal structure; the carbon skeleton and implicit-hydrogen conventions do not change. For example, ammonium (`[NH4+]`) has no nitrogen lone pair. On terminal hydrogen-bearing labels such as `OH`, `OH-`, and `NH2`, lone pairs are placed from the rendered heavy-atom glyph center. For charged bare atoms such as `[O-]` or `[Br-]`, the charge mark does not move the atom center used for lone pairs.

4 Atoms and charges

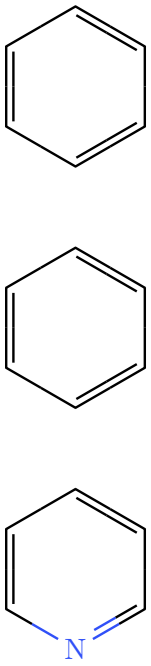
Standard SMILES syntax (atoms, bonds, branches, ring closures) is supported. This section covers the points specific to typed-smiles.

> **Note.** Ring closures may be written next to branch points. For example, `C1=CCCC(=O)1` closes the ring on the same atom that carries the C = O branch.

4.1 Aromatic rings

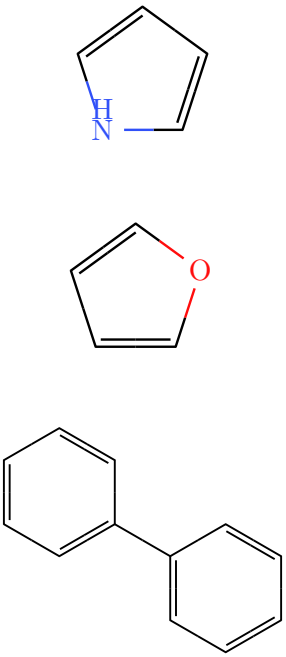
Aromatic rings can be written in either OpenSMILES form: lowercase aromatic atoms (`c1ccccc1`) or Kekulé notation with explicit alternating double bonds (`C1=CC=CC=C1`). Aromatic input is kekulized on parse, so both render identically.

```typst
#smiles("c1ccccc1") \
#smiles("C1=CC=CC=C1") \
#smiles("c1ccncc1")
```



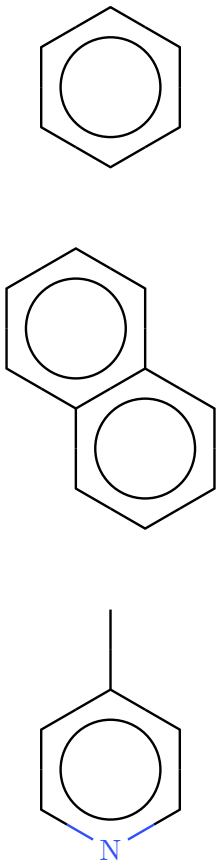
Bracket aromatic atoms carry their hydrogen count explicitly, as in pyrrole's `[nH]`. A single bond between two aromatic rings (biphenyl) may be written with or without the explicit `-`.

```typst
#smiles("c1cc[nH]c1") \
#smiles("c1occc1") \
#smiles("c1ccccc1-c1ccccc1", scale: 0.8)
```



`aromatic: "circle"` depicts lowercase-aromatic rings as single bonds with an inscribed circle instead of alternating double bonds. Each fully aromatic ring of a fused system gets its own circle; Kekulé-written input always keeps its explicit bonds.

```typst
#smiles("c1ccccc1", aromatic: "circle") \
#smiles("c1ccc2ccccc2c1", aromatic: "circle") \
#smiles("Cc1ccncc1", aromatic: "circle")
```

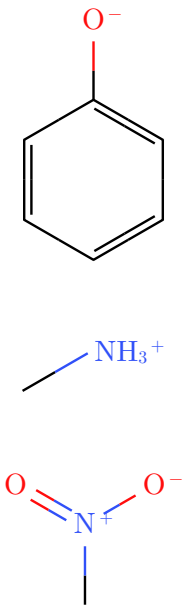


Note. Kekulization does not renumber anything: atom indices follow SMILES writing order for aromatic input exactly as for Kekulé input, so `show-indices`, `highlight(...)`, and `arrow(...)` references work unchanged, including on ring bonds whose double bond only appears after kekulization. A ring that cannot be kekulized (for example pyrrole written `c1ccnc1`, without the `[nH]` hydrogen) or an aromatic atom outside a ring is reported as an error.

4.2 Charges

Formal charges inside brackets render as a raised sign after the atom. Hydrogen counts and charge marks use the atom-label size so they remain legible when labels are scaled.

```typst
#smiles("[O-]C1=CC=CC=C1") \
#smiles("C[NH3+]") \
#smiles("C[N+](=O)[O-]")
```

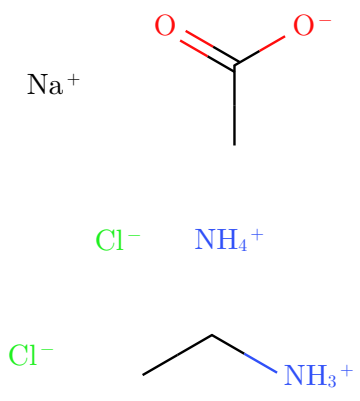


Note. A bracketed element such as `[N]` is a nitrogen atom. To print an arbitrary text label instead, use the abbreviation syntax `{N}` (see Section 8).

4.3 Salts and disconnected structures

A dot (`.`) means "no bond": it separates disconnected fragments, as in salts, counterions, and hydrates. Each fragment is laid out on its own and the fragments are arranged left to right in writing order.

```typst
#smiles("CC(=O)[O-].[Na+]") \
#smiles("[NH4+].[Cl-]") \
#smiles("CC[NH3+].[Cl-]")
```



Note. Atom indices stay global across fragments, still following SMILES writing order, so `show-indices`, `highlight(...)`, and `arrow(...)` can reference atoms in any fragment of the same `smiles()` call. A ring closure written across a dot (as in the OpenSMILES example `C1.C1`, ethane) still forms its bond.

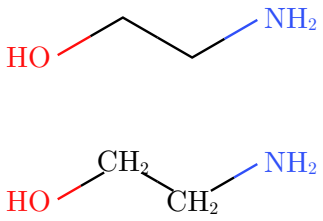
5 Hydrogens

5.1 Default display

Heteroatom hydrogens are shown by default; carbon hydrogens are hidden for a clean skeleton. `show-h: "all"` labels carbon hydrogens as well.

```
#smiles("OCCN") \
#smiles("OCCN", show-h: "all")
```

Typst

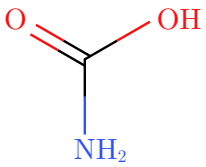


5.2 Explicit bracket hydrogens

A hydrogen written as its own bracket atom (for example the `[H]` atoms in `C([H])([H])[H]`) is folded into its neighbor's hydrogen count, exactly like an implicit hydrogen. Carbon-bound hydrogens stay hidden and heteroatom hydrogens are still labeled, so a fully hydrogen-suppressed SMILES depicts the same as its skeletal form. Hydrogen counts written inside a single bracket, such as `[OH-]` or `[NH4+]`, are unaffected and still shown.

```
#smiles("C([H])([H])([H])[H]") \
#smiles("N([H])([H])C(=O)O")
```

Typst

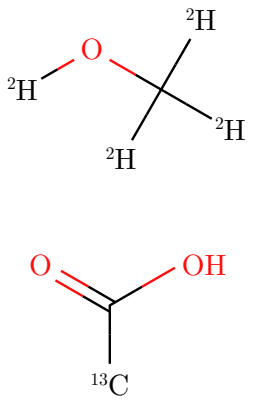


5.3 Isotopes

A bracket isotope is drawn as a leading superscript mass number. Isotopically labeled hydrogens such as deuterium `[2H]` are kept as drawn atoms rather than folded away.

```
#smiles("[2H]OC([2H])([2H])[2H]")
\
#smiles("[13CH3]C(=O)O")
```

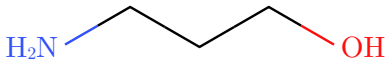
Typst



5.4 Label orientation

Terminal heteroatom labels orient so the bond meets the heavy atom rather than the trailing hydrogen, at any angle.

```typst
#smiles("NCCCO")
```



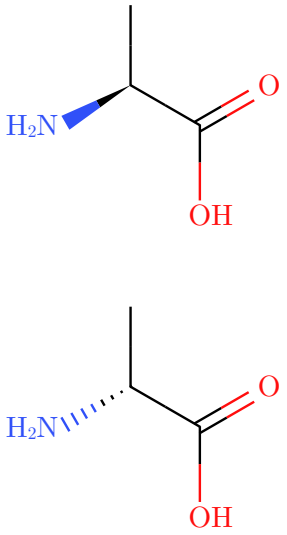
6 Stereochemistry

typed-smiles supports both standard SMILES stereo notations, plus a manual override for drawing wedges directly.

6.1 Tetrahedral centers: `@` and `@@`

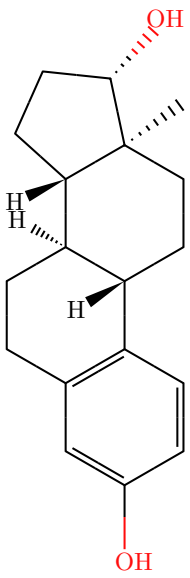
A bracket atom carrying `@` or `@@` becomes a wedge (toward the viewer) or hashed bond (away), computed from the 2D layout so the depiction matches the SMILES in the conventional orientation.

```typst
#smiles("N[C@@H](C)C(=O)O") \
#smiles("N[C@H](C)C(=O)O")
```



The renderer wedges an exocyclic substituent (such as OH or CH_3) where one exists, and an explicit hydrogen otherwise. Fused systems work too.

```typst
#smiles(
  "C[C@]12CC[C@H]3[C@H]
  ([C@@H]1CC[C@@H]2O)CCC4=C3C=CC(=C4)O",
  scale: 0.85,
)
```

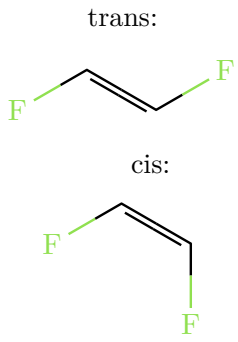


Note. The geometry is drawn correctly, but R/S descriptors are not computed. Ring stereochemistry between adjacent centers and bridged bicyclics may need a manual adjustment (see Section 17).

6.2 Double bonds: / and \

Directional bonds `/` and `\` around a double bond set its cis/trans geometry. They come in pairs, one on each side of the `=`. If an alkene carbon has two marked substituent bonds, the markers must place those substituents on opposite sides.

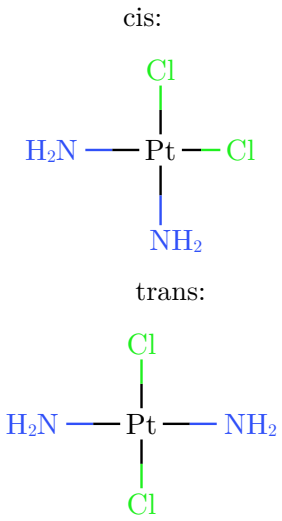
```
trans: #smiles("F/C=C/F")
#h(2em)
cis: #smiles("F/C=C\F")
```



6.3 Square-planar centers: @SP1, @SP2, @SP3

Square-planar stereochemistry is depicted exactly: the four neighbors sit at 90° around the center, arranged so the line traced through them in SMILES order forms the class's shape — 'U' for `@SP1`, '4' for `@SP2`, 'Z' for `@SP3`. The classic example is cis- versus trans-platin.

```
cis: #smiles("N[Pt@SP1](N)(Cl)Cl")
#h(2em)
trans: #smiles("N[Pt@SP2](N)(Cl)Cl")
```



Note. Trigonal-bipyramidal (`@TB1`–`@TB20`), octahedral (`@OH1`–`@OH30`), and allenal (`@AL1`, `@AL2`) centers are accepted and drawn with correct connectivity, but without stereo decoration — their 3D arrangement is not translated into wedges.

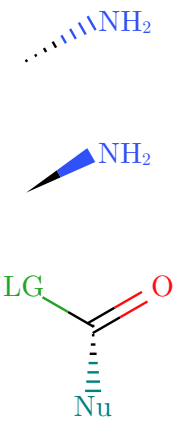
Note. Cumulated double bonds always label the central sp carbon (O = C = O, ketenes, allenes): the neighbors are collinear, so without the `C` the two double bonds would read as one long double bond.

6.4 Manual wedges: `!w` and `!h`

For direct control, `!w` draws a solid wedge and `!h` a hashed wedge. These are typed-smiles extensions, not standard SMILES. Like plain bonds, they are colored by the atoms at each end.

```
#smiles("C!wN") \
#smiles("C!hN") \
#smiles("C(!w{Nu|teal})({LG|green})=O")
```

Typst

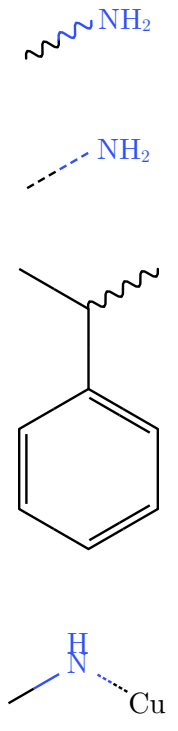


6.5 Wavy and dashed bonds: `!s` and `!d`

Two more drawing extensions on single bonds: `!s` draws a wavy (squiggly) bond — the conventional depiction of unspecified stereochemistry or an attachment point — and `!d` draws a dashed bond for hydrogen bonds, partial bonds, and coordination. Like plain bonds, they are colored by the atoms at each end.

```
#smiles("C!sN") \
#smiles("C!dN") \
#smiles("CC(!sC)C1=CC=CC=C1") \
#smiles("CN!d[Cu]")
```

Typst



> **Note.** `!s` and `!d` carry no cis/trans meaning and never consume a double-bond geometry slot; they can appear next to real `/` and `\` directional bonds.

> **Note.** Tip: Use empty custom labels `{}` or `!w{}` to draw bonds without a connecting atom.

7 Colors

7.1 Jmol CPK palette

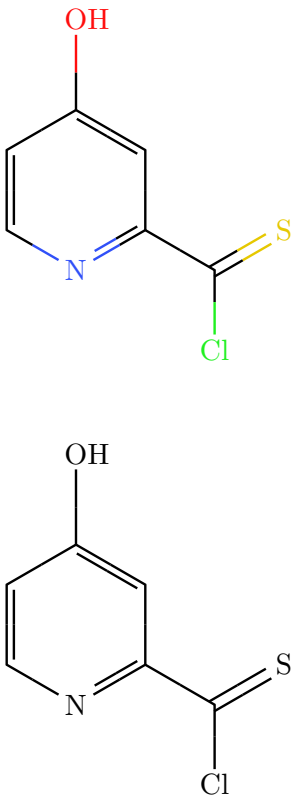
Atoms use the Jmol CPK palette by default. Carbon and unlisted elements are black; common heteroatoms get their conventional colors.

N	O	S	P	F	Cl	Br	I

Set `color: false` for a monochrome diagram.

```
#smiles("OC1=CC(=NC=C1)C(=S)Cl")
\
#smiles("OC1=CC(=NC=C1)C(=S)Cl", color:
false)
```

Typst



7.2 Custom colors (`atom-colors`)

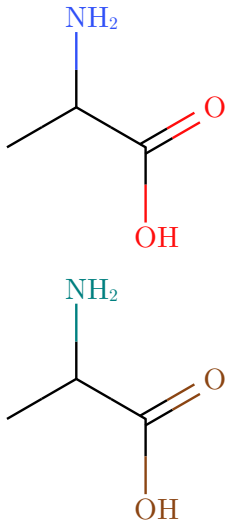
`atom-colors` is a single dictionary that overrides colors for both elements and arbitrary abbreviated groups. Any Typst color value works — `rgb()`, `luma()`, named constants, or anything else.

Element-symbol keys

Use an element symbol as the key to replace that element's CPK color everywhere it appears, including when referenced as an inline label style (`{OMe|O}` would pick up the overridden oxygen color).

Typst

```
// default CPK
#smiles("CC(N)C(=O)O")
// oxygen → dark brown; nitrogen → teal
#smiles("CC(N)C(=O)O",
  atom-colors: (O: rgb("#8B4513"), N: rgb("#008080")))
```

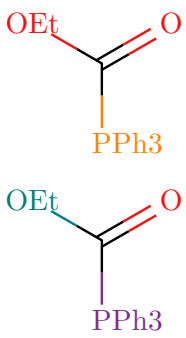


Label-name keys

Wrap the label text in braces to target a specific abbreviated group, regardless of any inline style it carries. The key is written as a quoted string because `{` is not a valid Typst identifier character.

Typst

```
// default — colors come from inline styles or element fallback
#smiles("{>PPh3|P}C({OEt|O})=O")
// override by label name — inline style is ignored for these
#smiles("{>PPh3|P}C({OEt|O})=O",
  atom-colors: ("{PPh3}": rgb("#7B2D8B"), "{OEt}": rgb("#008080")))
```

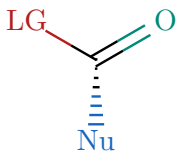


Mixing both key forms

Element keys and label keys can coexist in the same dictionary.

```
// O (element) → teal; {Nu} (label) → navy; {LG} (label) → maroon
#smiles("C(!w{Nu})({LG|red})=O",
  atom-colors: (O: rgb("#00897B"), "{Nu}": rgb("#1565C0"), "{LG}": rgb("#B71C1C")))
```

Typst



Priority

Color is resolved in this order for each atom or label:

1. Label-name key in `atom-colors` (e.g. `"{PPh3}"`), if the atom is an abbreviation.
2. Element-symbol key in `atom-colors` (e.g. `O`), for both real atoms and element-style labels.
3. The inline `{label|style}` style from the SMILES string (named color, hex, or element symbol).
4. The default CPK palette.

> **Note.** `color: false` is a hard override that sits above all of the above. When it is set, every atom and label renders in black regardless of any `atom-colors` entries or inline styles. If you want a mostly-monochrome diagram with one or two highlighted groups, keep or set `color: true` and put everything else in `atom-colors`.

7.3 Label colors (`{label|style}`)

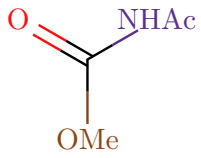
In the SMILES string, write `{label|style}` to color an abbreviated group independently of its element. The style can be a named color, an element symbol, or a hex code.

Named colors:

Name	Swatch	Name	Swatch
red		orange	
yellow		brown	
green		lime	
teal		cyan	
blue		navy	
purple		pink	
black		gray	
silver		maroon	
white			

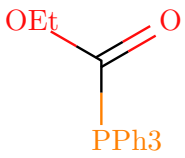
Hex colors: write a `#RRGGBB` hex code after the pipe to use any color. The `#` is just a regular character inside a Typst string literal.

```typst
#smiles("{OMe|#8B4513}C(=O){NHAc|#5B2E8C}")
```



Element symbol: using a symbol as the style applies that element's (possibly overridden) CPK color to the label and its bonds.

```typst
#smiles("{>PPh3|P}C({OEt|O})=O")
```



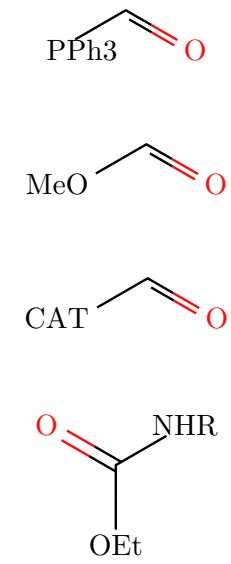
8 Abbreviated groups

8.1 Plain labels

Wrap text in braces `{...}` to place a labeled pseudo-atom that bonds like any other atom. Use `>` inside the label to choose the attachment glyph; the marker is not displayed. Rotate the molecule when needed so the bond approaches the chosen glyph roughly perpendicular to the written label.

```
#smiles("{>PPh3}C=O") \
#smiles("{Me>O}C=O") \
#smiles("{CA>T}C=O") \
#smiles("{OEt}C(=O){NHR}")
```

Typst

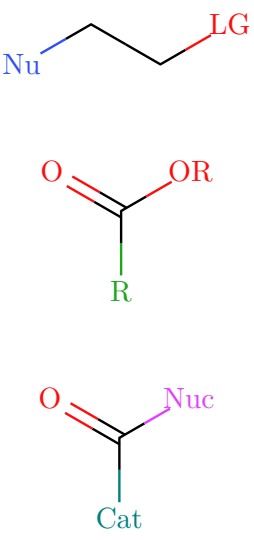


8.2 Colored labels

Add `|style` inside the braces to color a label. Use a named color, an element symbol, or a hex code — see Section 7 for the full list.

```
#smiles("{Nu|blue}CC{LG|red}") \
#smiles("{R|green}C(=O){OR|O}") \
#smiles("{Cat|teal}C(=O){Nuc|#E040FB}")
```

Typst

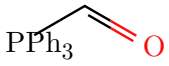


8.3 Scripted labels

Use `_(...)`, `^(...)`, or their one-character forms `_4` and `^+` for explicit subscripts and superscripts in a label. When both follow one glyph, they are paired with that glyph: `{NH_4^+}` places the plus above H rather than above 4. Script size, spacing, and vertical placement match `ce()` notation. Parenthesize multi-character scripts.

```
#smiles("{>PPh_(3)}C=O") \
#smiles("{NH_4^+}") \
#smiles("{SO_(4)^(2-)}")
```

Typst



9 Chemical formulas: `ce()`

For formulas and text equations, `ce` is re-exported from the chemformula package, so one import covers both structures and formulas.

```
#ce("H2SO4") #h(1em) #ce("Ca^2+")
#h(1em) #ce("2 H2 + O2 -> 2 H2O")
```

Typst

$$\mathrm{H_2SO_4} \quad \mathrm{Ca^{2+}} \quad \mathrm{2\,H_2 + O_2 \longrightarrow 2\,H_2O}$$

9.1 Fonts and size

`ce` accepts `font` and `font-size` any other arguments pass through to chemformula.

```
#ce("CuSO4 * 5 H2O") \
#ce("CuSO4 * 5 H2O", font: "Libertinus
Serif", font-size: 13pt) \
#ce("CuSO4 * 5 H2O", font: "PT Sans",
font-size: 13pt)
```

Typst

$$\mathrm{CuSO_4 \cdot 5\,H_2O}$$
$$\mathrm{CuSO_4 \cdot 5\,H_2O}$$
$$\mathrm{CuSO_4 \cdot 5\,H_2O}$$

10 Molecular weight: `mol-weight()`

`mol-weight(smiles)` returns the molecular weight in g/mol as a plain `float`: the sum of IUPAC standard atomic weights over every atom, including implicit and explicit hydrogens. Round it for display with `calc.round`.

```typst
Ethanol: #calc.round(mol-weight("CCO"), digits: 2) g/mol \
Caffeine: #calc.round(
  mol-weight("CN1C=NC2=C1C(=O)N(C(=O)N2C)C"),
  digits: 2,
) g/mol \
Sodium acetate: #calc.round(
  mol-weight("CC(=O)[O-].[Na+]"),
  digits: 2,
) g/mol
```

Typst

Ethanol: 46.07 g/mol
Caffeine: 194.19 g/mol
Sodium acetate: 82.03 g/mol

Dot-separated fragments are summed together, so salts and hydrates work as written. Charges are accepted; as in standard formula-weight arithmetic, the electron mass difference of ions is ignored.

> **Note.** Compilation fails with a descriptive error when the weight is undefined: wildcard `*` atoms, `{label}` abbreviations (no defined composition), and isotope-labeled atoms such as `[2H]` (a specific nuclide needs a nuclide mass, not a standard atomic weight).

11 Inline molecules: `smiles-inline()`

`smiles-inline(smiles-str, height: 1.4em, baseline: auto, ..args)` scales a structure to a target height and baseline-aligns it so it reads inline, like a word in the sentence. Extra named arguments are passed straight to `smiles()`.

```
Dehydrating ethanol #smiles-inline("CCO")
over alumina gives ethylene
#smiles-inline("C=C"); toluene
#smiles-inline("Cc1ccccc1") is a common
solvent.
```

Typst

Dehydrating ethanol  over alumina gives ethylene  toluene  is a common solvent.

Sizing works in two steps: the drawing is scaled to fit the `height` (default `1.4em`, so ordinary line spacing stays essentially unchanged), and the scale is capped so one bond never exceeds most of that height — otherwise a flat, mostly horizontal molecule, which measures almost no height, would blow up to fill it. Typst grows a line to fit tall inline content, so neighboring lines are never overlapped; passing a larger `height` only makes the molecule's own line taller.

> **Note.** `baseline` sets how far the drawing's vertical center sits above the text baseline (default centers it on the lowercase body). Atom labels scale down with the drawing: at inline sizes, label-heavy molecules read better with a larger `height` or as a display-mode `smiles()`.

12 CeTZ integration: `smiles-cetz()`

`smiles-cetz(smiles-str, name:, origin:, fg:, theme:, ..opts)` draws a molecule **inside an existing CeTZ canvas** and registers named anchors on the group, so arbitrary CeTZ drawing attaches to real molecular positions:

- `"<name>.atom-<i>"` — every atom center (writing-order indices, as shown by `show-indices`),
- `"<name>.bond-<i>-<j>"` — every bond midpoint (with $i < j$),
- `"<name>.center"` — the molecule origin.

Coordinates are in bond-length units; give the canvas `length: 30pt * scale` so sizes match `smiles(scale: ...)`, and wrap the canvas in `context` (atom-label measurement needs it — plain `smiles()` provides it automatically, a raw canvas does not).

The Watson–Crick A–T base pair below is two `smiles-cetz` calls plus plain CeTZ: two dashed hydrogen bonds drawn between atom anchors of the two bases, endpoint nudges so the dashed lines do not start at atom centers, distance labels, and glycosidic-bond stubs to the sugar backbone.

Typst

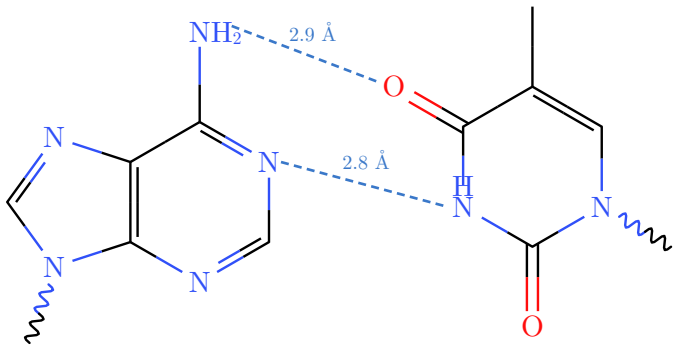
```
#align(center, context cetz.canvas(length: 30pt, {
import cetz.draw: *

smiles-cetz("Nc1ncnc2N(!s{})cnc12", name: "A")
smiles-cetz("Cc1cN(!s{})c(=O)[nH]c1=O", name: "T", origin: (4.9, 0.42))

let hb = (paint: rgb("#3A78C9"), thickness: 1.0pt, dash: "densely-dashed")
let off(anchor, by) = (rel: by, to: anchor)
line(off("A.atom-11", (0.4, -0.15)), off("T.atom-9", (-0.2, 0.06)), stroke: hb)
line(off("A.atom-2", (0.15, 0)), off("T.atom-7", (-0.2, 0)), stroke: hb)

// Hydrogen-bond distances.
content((rel: (0.2, 0.2), to: ("A.atom-11", 50%, "T.atom-9")), text(size: 7.5pt, fill: rgb("#3A78C9"))[2.9 Å])
content((rel: (0, 0.28), to: ("A.atom-2", 50%, "T.atom-7")), text(size: 7.5pt, fill: rgb("#3A78C9"))[2.8 Å])

}))
```



The remaining `..opts` are `smiles()` drawing options (`color`, `rotation`, `mirror`, `show-h`, `aromatic`, `opacity`, `bond-customizations`, …). `fg` and `theme` default to black and light because `auto` foreground resolution is not available inside a raw canvas — pass them explicitly on dark backgrounds.

Use regular CeTZ relative coordinates to offset any connection endpoint: `(rel: (0.12, 0.04), to: "A.atom-0")`. This is the same idea as `atom(..., offset:)` in mechanism arrows, but in raw CeTZ syntax.

> **Note.** Turn on `show-indices: true` while writing anchor references, exactly as for mechanism arrows, to read off the atom indices. Bond-midpoint anchors (`bond-<i>-<j>`) attach to the middle of a bond — useful for marking a reacting or attachment bond.

13 Reaction schemes

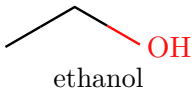
Three helpers compose molecules, formulas, and arrows into schemes.

13.1 mol()

`mol(spec, label: none, offset: (0,0), ..opts)` is a reaction item. `spec` is either any content (`smiles(...)`, `ce(...)`, text) or a SMILES **string**. Passing a string lets `reaction` render the molecule itself so its atoms become addressable by curly arrows (see Reaction mechanisms). String molecules accept common drawing options such as `scale`, `font-size`, `font`, `bond-stroke`, `color`, `rotation`, `show-h`, `lone-pairs`, `opacity`, `bond-customizations`, `atom-colors`, and `show-indices`. `offset` nudges the molecule in page coordinates, in bond-length units: positive x moves right and positive y moves up regardless of `reaction(flow:)`. In mechanism mode, `reaction(scale: ...)` sets the shared canvas scale (and its offsets), and each `mol(scale: ...)` multiplies it for that molecule alone. A `mol()` item can also be placed in a `rxn-arrow(above:/below:)` slot to draw a molecule over or under the arrow. In ordinary schemes, an offset also changes the reaction's layout bounds, so `brackets()` encloses the shifted edge.

```
1  #reaction(mol(smiles("CCO"),
   label: [ethanol]))
```

Typst



ethanol

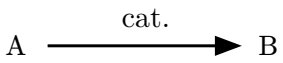
13.2 rxn-arrow()

`rxn-arrow(above: none, below: none, dir: auto, kind: "single", scale: 1.0)` draws an arrow. `dir` may be `"right"`, `"left"`, `"up"`, `"down"`, or `auto` (the default, following the enclosing `reaction(flow:)`). `kind` may be `"single"`, `"equilibrium"`, `"equilibrium-filled"`, `"dashed"`, or `"wavy"`. `above` and `below` carry reagents and conditions — any content, or a `mol()` item to draw a molecule in the label slot (in mechanism mode a `mol()` there also becomes a referenceable species; see Reaction mechanisms). `scale` uniformly resizes the complete arrow, including its condition labels. `color` defaults to `auto`, inheriting the surrounding text color (so arrows recolor on dark slides).

Argument	Default	Effect
`above`	`none`	Label above a horizontal arrow, or right of a vertical arrow. Content or a `mol()` item.
`below`	`none`	Label below a horizontal arrow, or left of a vertical arrow. Content or a `mol()` item.
`dir`	`auto`	Arrow direction: `"right"`, `"left"`, `"up"`, `"down"`, or `auto` (follows `reaction(flow:)`).
`kind`	`"single"`	Arrow style: `"single"`, `"equilibrium"`, `"equilibrium-filled"`, `"dashed"`, or `"wavy"`.
`scale`	`1.0`	Uniform arrow scale, including condition labels.
`color`	`auto`	Arrow color; `auto` inherits the surrounding text color.

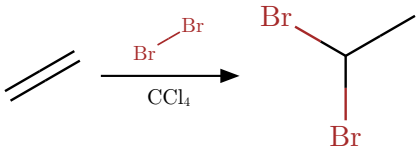
Scale an individual arrow without changing the rest of its reaction. The shaft, arrowhead, spacing, and condition labels resize together.

```typst
#reaction(
  ce("A"),
  rxn-arrow(scale: 1.4, above: [cat.]),
  ce("B"),
)
```



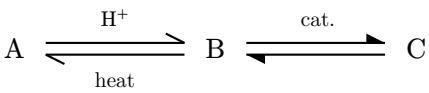
An arrow label slot also accepts a `mol()` item, drawing a molecule above or below the arrow with its own per-molecule options.

```typst
#reaction(
  mol("C=C"),
  rxn-arrow(above: mol("BrBr", scale: 0.7), below: [CCl#sub[4]]),
  mol("BrC(Br)C"),
)
```



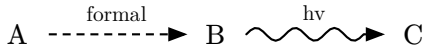
Equilibrium arrows use paired half-heads. The filled variant uses filled half-heads.

```typst
#reaction(
  ce("A"),
  rxn-arrow(kind: "equilibrium", above: ce("H+"), below: [heat]),
  ce("B"),
  rxn-arrow(kind: "equilibrium-filled", above: [cat.]),
  ce("C"),
)
```



A dashed arrow marks a hypothetical or formal transformation; a wavy arrow marks a distorted or non-elementary one.

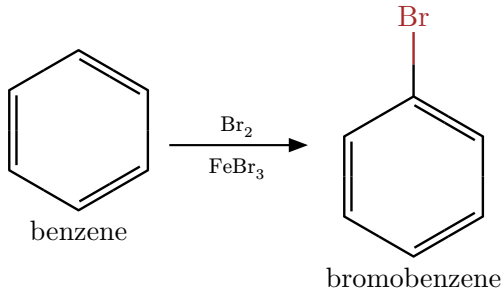
```typst
#reaction(
  ce("A"),
  rxn-arrow(kind: "dashed", above: [formal]),
  ce("B"),
  rxn-arrow(kind: "wavy", above: [hv]),
  ce("C"),
)
```



13.3 `reaction()`

`reaction(gap-h: 1.5em, gap-v: 1.5em, scale: 1.0, breakable: false, show-indices: false, flow: "right", ..items)` lays out molecules and arrows left to right. An up or down arrow wraps the scheme onto a new row.

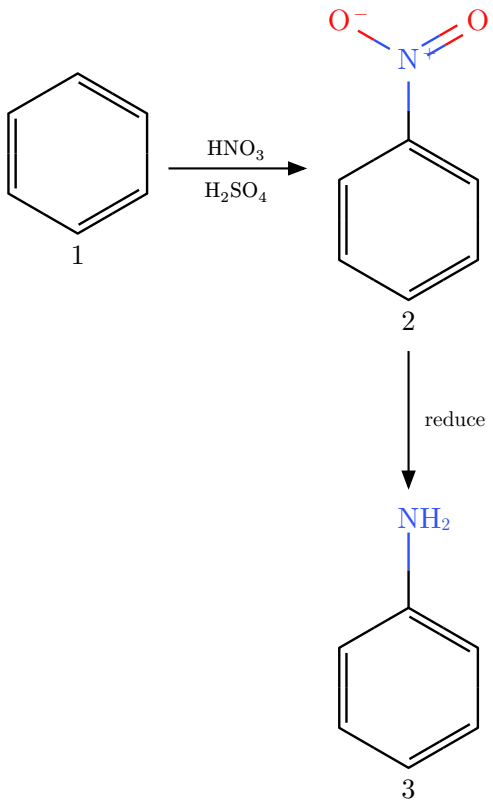
```typst
#reaction(
  mol(smiles("C1=CC=CC=C1"), label: [benzene]),
  rxn-arrow(above: ce("Br2"), below: ce("FeBr3")),
  mol(smiles("BrC1=CC=CC=C1"), label: [bromobenzene]),
)
```



Note. Set `reaction(show-indices: true)` to stamp indices on every string SMILES molecule rendered through `mol("...")` in that reaction. This is useful while authoring large mechanisms with many arrows or highlights. A molecule can still opt out with `mol("...", show-indices: false)`. Molecules already pre-rendered as content, such as `mol(smiles("..."))`, keep their own `smiles()` options.

A wrap-around scheme using a downward arrow:

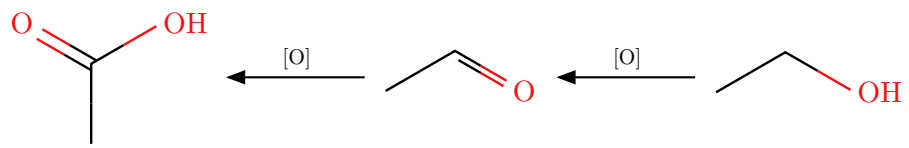
```typst
#reaction(
  mol(smiles("C1=CC=CC=C1"), label: [1]),
  rxn-arrow(above: ce("HNO3"), below: ce("H2SO4")),
  mol(smiles("[O-][N+](=O)C1=CC=CC=C1"), label: [2]),
  rxn-arrow(dir: "down", above: [reduce]),
  mol(smiles("NC1=CC=CC=C1"), label: [3]),
)
```



Flow direction

`flow` sets the writing direction: `"right"` (default), `"left"`, `"up"`, or `"down"`. `"left"` and `"up"` reflect the scheme along the flow axis and flip the arrowheads, so a scheme written in reading order displays right-to-left or bottom-to-top. Ordinary non-arrow items follow the same direction, so `flow: "up"` and `"down"` stack molecules, plus signs, and text vertically even when no `rxn-arrow()` separators are present. This is the natural choice for a branch that grows out of the left or bottom of a `cycle()` — its released product ends up on the side facing the cycle. Arrows created with `rxn-arrow(dir: auto)` (the default) follow the flow; an explicit `dir` is kept.

```typst
#reaction(flow: "left",
  mol(smiles("CCO")),
  rxn-arrow(above: [\[O\]]),
  mol(smiles("CC=O")),
  rxn-arrow(above: [\[O\]]),
  mol(smiles("CC(=O)O")),
)
```

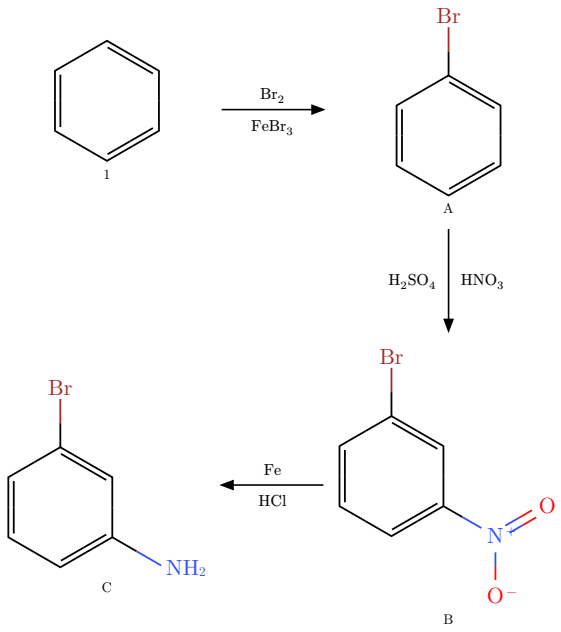


Note. Because `step(out:)` of a `cycle()` accepts any content, a nested `reaction(flow: "left")` (or `"down"`) grows a full sub-reaction out of a released molecule that reads naturally away from the ring, while the outer reaction continues in its own direction. Flow applies to the scheme layout; mechanism mode (curly arrows) is unaffected.

Uniform scale

Pass `scale` to shrink or enlarge the entire scheme uniformly — molecules, arrows, labels, and separators all change together. This is especially useful for multi-step or wrap-around schemes that would otherwise be too wide.

```typst
// same scheme at three different scales
#reaction(
  scale: 0.75,
  mol(smiles("C1=CC=CC=C1"), label: text(size: 7pt)[1]),
  rxn-arrow(above: ce("Br2"), below: ce("FeBr3")),
  mol(smiles("BrC1=CC=CC=C1"), label: text(size: 7pt)[A]),
  rxn-arrow(dir: "down", above: ce("HNO3"), below: ce("H2SO4")),
  mol(smiles("BrC1=CC(=CC=C1)[N+](=O)[O-]"), label: text(size: 7pt)[B]),
  rxn-arrow(dir: "left", above: ce("Fe"), below: ce("HCl")),
  mol(smiles("BrC1=CC(=CC=C1)N"), label: text(size: 7pt)[C]),
)
```



Note. `reaction(scale: ...)` and `smiles(scale: ...)` are independent. The reaction scale is applied on top of however each individual molecule is sized. To resize everything from a single place, use `reaction(scale: ...)` and let each `smiles()` call use its default.

Page-break behaviour

By default, `reaction` sets `breakable: false`, so the whole scheme moves to the next page as a unit if it does not fit on the current one. This prevents a molecule or vertical branch from stranding on a different page from the rest of the scheme. Set `breakable: true` only for very long schemes that must span pages.

Typst

```typst
// Default — the scheme always stays on one page:
#reaction( … )                      // breakable: false

// Opt in to page splitting for very long schemes:
#reaction(breakable: true, … )
```

14 Reaction mechanisms

`reaction()` also draws electron-pushing mechanisms: curly arrows that flow from a lone pair, bond, or atom into another bond or atom — within one structure or across separate species. It switches from grid layout to a single shared canvas as soon as any curly `arrow()` or `highlight()` is present. `mol(offset:)` by itself stays in scheme layout and nudges the species inside its grid cell.

14.1 Referencing atoms

Atoms are addressed by their **writing-order index** (0-based), so the SMILES string is never modified. Inside `smiles()` use single-index references; inside `reaction()` prefix them with the species index `s` — the position of the `mol()`/content item in written order. `mol()` items inside `rxn-arrow(above:/below:)` slots are counted too, at the arrow's position in the sequence (`above` before `below`); the arrows themselves, plain arrow-label content, and annotations are not counted.

Reference	Resolves to
`atom(i)` / `atom(s, i)`	Atom center.
`bond(i, j)` / `bond(s, i, j)`	Midpoint of the bond between atoms i and j.
`lp(i)` / `lp(s, i)`	A lone pair on atom i (use `pair: n` to pick one).
`species(k)`	Bounding-box edge of a whole item (e.g. a `ce()` formula).

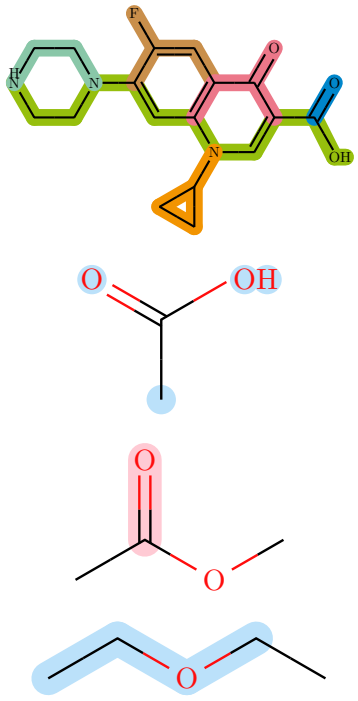
Note. Every reference accepts an `offset: (dx, dy)` nudge in bond-length units — the escape hatch for fine-tuning an endpoint or pointing at a single-electron dot. Set `show-indices: true` on a `smiles()` / `mol()` to stamp the indices on the diagram while you write the arrows.

Hydrogen atoms declared inside bracket notation (e.g. `[OH-]`, `[NH4+]`) are also addressable. Each H gets the next available index after all heavy atoms, so `atom(1)` in `[OH-]` refers to the hydrogen. With `show-indices`, the heavy-atom and H badges are centered on their rendered label glyphs. Charge marks are excluded from the atom center, so `atom(0)` in `[O-]` points to the O glyph rather than the combined `O-` label.

14.2 `arrow()` and `highlight()`

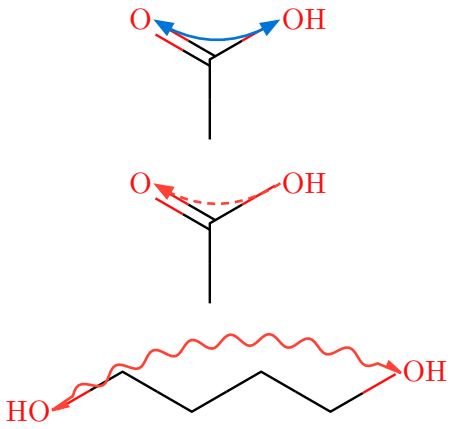
`arrow(from:, to:, label: none, color: red, bend: "left", angle: 15deg, half: false, heads: "end", style: "solid")` draws a curly arrow between two references. `bend` is `"left"`, `"right"`, or `none` `angle` sets how strongly it bows; `half: true` draws fishhook (single-electron) heads. `heads` selects which ends carry an arrowhead — `"end"` (default), `"both"`, or `"none"` — and `style` the shaft: `"solid"` (default), `"dashed"`, or `"wavy"` both combine freely with `bend`. `highlight(ref, fill:)` shades an atom (disk) or bond (capsule) behind the structure. Pass an array of references to shade several atoms or bonds with one call. Bond highlights are trimmed away from atom centers by default; set `include-atoms: true` to also shade the endpoint atoms of each highlighted bond. Abbreviated group labels such as `{PPh3}` are highlighted as measured label-width capsules rather than atom-sized disks.

```
#smiles(
  "N1CCN(CC1)C(C(F)=C2)=CC(=C2C4=O)N(C3CC3)
  highlight((bond(0, 5), bond(5, 4),
  bond(4, 3), bond(3, 6), bond(6, 10),
  bond(10, 11), bond(11, 15), bond(15,
  19), bond(19, 20), bond(20, 21),
  bond(21, 23), bond(23, 25)), fill:
  rgb(150, 191, 13), include-atoms:
  true),
  highlight((bond(15, 16), bond(16, 18),
  bond(18, 17), bond(17, 16)), fill:
  rgb(242, 148, 1), include-atoms: true),
  highlight((bond(3, 2), bond(2, 1),
  bond(1, 0)), fill: rgb(137, 199, 168),
  include-atoms: true),
  highlight((bond(6, 7), bond(7, 8),
  bond(7, 9), bond(9, 12)), fill:
  rgb(201, 143, 75), include-atoms:
  true),
  highlight((bond(11, 12), bond(12, 13),
  bond(13, 20), bond(13, 14)), fill:
  rgb(236, 119, 137), include-atoms:
  true),
  highlight((bond(21, 22)), fill: rgb(0,
  134, 203), include-atoms: true),
  color: false,
  rotation: 90deg,
  bond-stroke: 0.8pt,
  scale: 0.5,
)
#smiles(
  "CC(=O)O",
  highlight((atom(0), atom(2), atom(3),
  atom(4)), fill: rgb("#BBE1FA")),
)
#smiles(
  "CC(=O)OC",
  highlight((bond(2, 1)), include-
  atoms:true, fill: rgb("#FFCAD4")),
)
#smiles(
  "CCOCC",
  highlight(
    (bond(0, 1), bond(1, 2), bond(2, 3)),
    fill: rgb("#BBE1FA"),
    include-atoms: true,
  ),
```



`heads` and `style` tune the arrow shape — for instance a double-headed dashed arrow for an interaction, or a wavy arrow for a photochemical or homolytic step.

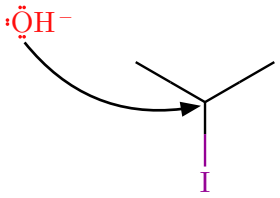
```typst
#smiles("CC(=O)O",
  arrow(from: atom(3), to: atom(2), heads: "both", color: blue))
#smiles("CC(=O)O",
  arrow(from: atom(3), to: atom(2), style: "dashed"))
#smiles("OCCCCO",
  arrow(from: atom(0), to: atom(5), style: "wavy", heads: "both", half: true))
```



14.3 A mechanism across species

Hydroxide attacks the central carbon; the C–I bond breaks toward iodide. The nucleophile is offset so the curly arrow reads cleanly.

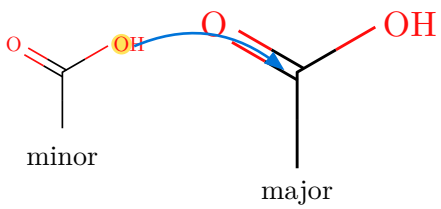
```typst
#reaction(
  mol("[OH-]", lone-pairs: "dots", offset: (0, 1)),
  mol("C(I)(C)C"),
  arrow(from: lp(0, 0, offset:(0.1, -0.2)), to: atom(1, 0, offset : (0.1, -0.1)),
        bend: "right", color : black),
)
```



14.4 Per-molecule scale in mechanisms

In mechanism mode `reaction(scale:)` sets the shared canvas scale and each `mol(scale:)` multiplies it for its own species. Curly arrows and highlights land on the scaled coordinates.

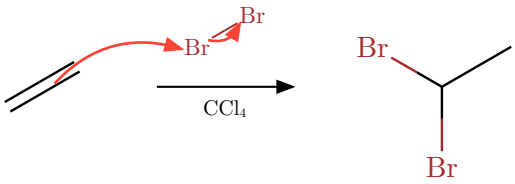
```typst
#reaction(
  mol("CC(=O)O", scale: 0.7, label: [minor]),
  mol("CC(=O)O", scale: 1.2, label: [major]),
  highlight(atom(0, 3)),
  arrow(from: atom(0, 3), to: atom(1, 1), color: blue),
)
```



14.5 Referencing molecules above an arrow

A `mol()` placed in a `rxn-arrow(above:/below:)` slot is lifted onto the shared canvas as a species of its own, so its atoms take part in the mechanism like any other species.

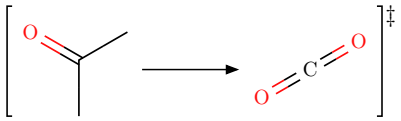
```typst
#reaction(
  mol("C=C"),
  rxn-arrow(above: mol("BrBr", scale: 0.8), below: [CCl#sub[4]]),
  mol("BrC(Br)C"),
  arrow(from: bond(0, 0, 1), to: atom(1, 0), color: red),
  arrow(from: atom(1, 0), to: atom(1, 1), color: red, bend: "right"),
)
```



14.6 `brackets()`

`brackets(body, sup: none, sub: none)` encloses any content in square brackets including another reaction, with optional marks at the corners — a `[‡]` for a transition state or a charge for a reactive intermediate.

```typst
#brackets(
  [#reaction(smiles("CC(=O)C"), rxn-arrow(), smiles("O=C=O"), scale : 0.7)],
  sup: [‡])
```

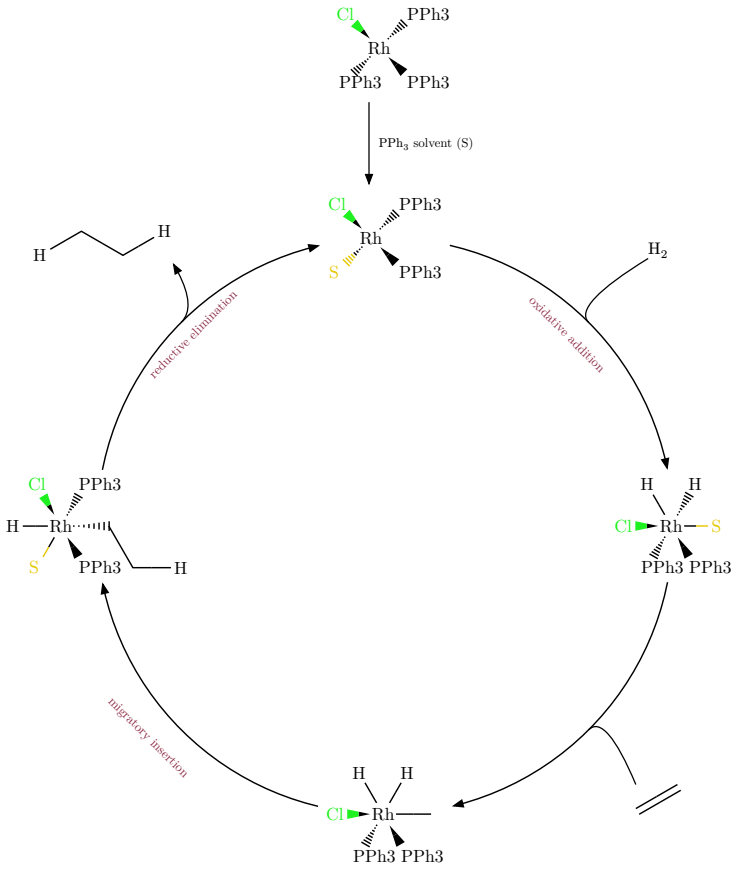


15 Catalytic cycles: `cycle()`

`cycle(..items, radius:, start:, clockwise:, scale:, reagent-bend:, ...)` arranges species on a circle with arc arrows between them — the standard depiction of a catalytic cycle. Items alternate species and step()s, just as `reaction()` alternates molecules and arrows, but the sequence closes into a ring (the last step returns to the first species). Species are mol() items or any content; SMILES strings are rendered by `smiles()`.

`step(label:, into:, out:, bend:, merge:, rotation:, label-offset:, into-offset:, out-offset:)` describes the arc that follows a species: `label` names the transformation, `into` adds a reagent merging into the arc from outside the ring, and `out` a product leaving it. Those side arrows are bookkeeping arrows for material entering or leaving the catalytic manifold, not electron-pushing arrows. `merge: true` draws a side arrow tangent to the arc so it visually fuses with the main cycle arrow. `rotation` rotates the step label: `"straight"` keeps it horizontal, `"auto"` follows the step's circle angle while keeping text upright, and an explicit angle is used as given. The three `*-offset` arguments nudge the label and the into/out reagents like a `mol` offset.

```typst
#align(center)[
#reaction(
  scale : 0.6,
  flow : "down",
  mol(smiles("{Rh}(!w{>PPh3})(!h{>PPh3})(!hCl)(!w{>PPh3})"), offset: (0.5, 0)),
  rxn-arrow(above : [#ce("PPh3") solvent (S)]),
  cycle(
    radius : 6.0,
    mol(smiles("{Rh}(!w{S | S})(!h{>PPh3})(!hCl)(!w{>PPh3})")),
    step(label : [#text(size : 8pt)[oxidative addition]], label-offset : (0.3, 0.5),
    into : [#ce("H2")], rotation : "auto", merge : true),
    mol(smiles("{Rh}(!w{>PPh3})(!h{>PPh3})(!hCl)({S | S})({H})(!w{H})")),
    step(into : [#smiles("C=C")], merge : true),
    mol(smiles("{Rh}(!w{>PPh3})(!h{>PPh3})(!hCl)({})({H})({H})")),
    step(label : [#text(size : 8pt)[migratory insertion]], label-offset: (0, 0.5),
    rotation : "auto"),
    mol(smiles("{Rh}(!w{>PPh3})(!h{>PPh3})(!hCl)({S | S})({H})(!wCC{H})", rotation :
    -90deg)),
    step(label : [#text(size : 8pt)[reductive elimination]], label-offset:  (-0.5,
    0.5), out : [#smiles("{H}CC{H}")], merge:true, rotation : "auto")
  )
)
]
```

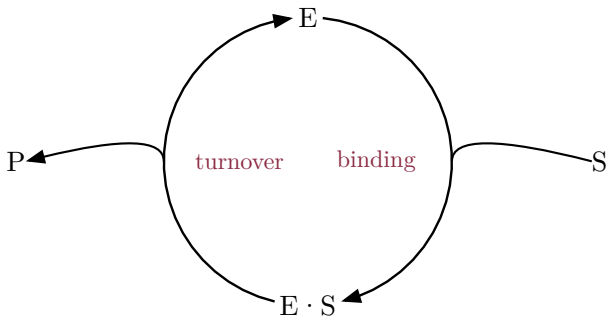


Because `step(out:)` accepts any content, passing a `reaction()` there grows a full linear branch out of a released molecule — the released product becomes the first species of a downstream sequence. Reagents (both `into` and `out`) are offset by their own measured size, so even a wide nested reaction clears the ring. Incoming content is attached on the side facing the cycle so the side arrow starts at the last upstream item; outgoing content uses the same side so the leaving arrow lands at the first downstream item instead of the center of the whole branch.

> **Note.** `radius` defaults to `auto`, which fits the species without overlap; set it explicitly for a tighter or looser ring. `start` is the angle of the first species (`90deg` is the top) and `clockwise` the travel direction. `scale` sets the canvas bond length as in `reaction()`. `arc-gap` tunes how close the arc arrows sit to the species (in bond-length units beyond each molecule's radius; smaller or even negative brings them closer). `reagent-bend` sets the default curvature of into/out side arrows; `step(bend:)` overrides one step, and `bend: 0` makes that side arrow nearly radial. A one-species `cycle()` draws that species alone.

With `merge: true` the into/out arrows fuse tangentially with the main cycle arrows, and `arc-gap` pulls the arrows closer to the species:

```typst
#cycle(
  scale: 0.9,
  arc-gap: 0.0,
  mol(box(inset: 2pt)[E]),
  step(label: [binding], into: [S], merge: true),
  mol(box(inset: 2pt)[E·S]),
  step(label: [turnover], out: [P], merge: true),
)
```



16 Project-wide defaults

Typst functions support partial application with `.with()`. Calling `smiles.with(...)` returns a new function with the given arguments pre-filled. Rebind the name in your preamble and every subsequent `#smiles(...)` call uses those defaults — including inside `#reaction()`, because the content is evaluated before `reaction` sees it.

Any argument can be pre-filled this way: `bond-length`, `font`, `scale`, `font-size`, `atom-colors`, or any other `smiles` parameter.

Typst

```typst
// ── preamble ──────────────────────────────────────────────────────
#import "@preview/typed-smiles:0.7.0": *

#let smiles = smiles.with(
  bond-length: 0.9,
  font:        "Libertinus Serif",
  atom-colors: (O: rgb("#8B4513"), N: rgb("#008080")),
)

// ── anywhere in the document ──────────────────────────────────────
#smiles("CC(N)C(=O)O")   // uses bond-length 0.9, Libertinus, custom O/N
#reaction(
  smiles("CCO"),
  rxn-arrow(),
  smiles("CC(=O)O"),     // same custom smiles — preamble applies here too
)
```

Note. Any argument passed directly at the call site still overrides the `.with()` default for that call alone.

17 Limitations

- Directional bonds (`/`, `\`) draw the correct cis/trans geometry, but E/Z descriptors are not computed.
- R/S and E/Z descriptors are not calculated.
- Trigonal-bipyramidal (`@TB`), octahedral (`@OH`), and allenal (`@AL`) centers are accepted but drawn without stereo wedges.
- Ring stereochemistry between adjacent centers and bridged bicyclics can overlap or need a manual adjustment (try `rotation`, or the manual `!w` and `!h` wedges). Ordinary substituent branches resolve crowding automatically — a blocking branch flips to the other side of its attachment bond (e.g. the ortho acetyl and carboxyl groups of aspirin), or the colliding bond flips its zigzag turn — always keeping ideal bond angles, so plain branches do not overlap.

18 Quick reference

18.1 Syntax

Syntax	Meaning
`c n o …`	Aromatic atom (lowercase); kekulized on parse.
`.`	Disconnected fragment (no bond): salts, hydrates.
`[...]`	Bracket atom (any element, charges, explicit H).
`@` / `@@`	Tetrahedral anticlockwise / clockwise.
`@SP1`–`@SP3`	Square planar (U / 4 / Z shape), depicted exactly.
`@TB` `@OH` `@AL`	Extended stereo: accepted, drawn without wedges.
`$`	Quadruple bond (four parallel lines).
`/ \`	Double-bond cis/trans geometry.
`!w` / `!h`	Manual solid / hashed wedge (typed-smiles extension).
`!s` / `!d`	Wavy / dashed bond (typed-smiles extension).
`{label}`	Abbreviated group pseudo-atom.
`{>label}`	Abbreviated group anchored at the glyph after `>`.
`{label|style}`	Colored abbreviated group.

18.2 `smiles()` options

Option	Default	Effect
`style`	`"default"`	Journal preset: `"acs"`, `"rsc"`, `"nature"`, `"wiley"`.
`scale`	`1.0`	Balanced size of everything.
`bond-length`	`none`	Bond length only (`1.0` is 30 pt).
`font-size`	`none`	Atom-label size only.
`font`	`"New Computer Modern"`	Atom-label font.
`bond-stroke`	`none`	Bond width only.
`color`	`auto`	CPK colors for `"default"` monochrome for journal presets.
`fg`	`auto`	Foreground for bonds/carbon labels; `auto` inherits the text color.
`theme`	`auto`	CPK palette variant; `auto` goes dark when `fg` is light.
`rotation`	`0deg`	Rotate, labels stay upright.
`mirror`	`none`	Optional `"horizontal"` or `"vertical"` reflection.
`show-h`	`()`	Label selected implicit hydrogens; use `"all"` for every atom.

`aromatic`	`"kekule"`	Lowercase-aromatic rings as doubles or `"circle"`.
`atom-annotations`	`()`	Small gray side labels as `(index, content)` or `(index, content, offset)` tuples.
`opacity`	`100%`	Fade the whole drawing (ghost molecules).
`bond-customizations`	`()`	Per-bond `color`, `stroke`, `opacity` keyed by `bond(i, j)`.
`lone-pairs`	`none`	Draw lone pairs as `"dots"` or `"lines"`.
`atom-colors`	`(:)`	Color overrides: `O: red` for elements, `"{PPh3}": blue` for labels.
`show-indices`	`false`	Stamp atom indices for writing references.
`..annotations`	—	`arrow()` / `highlight()` items on this molecule.

18.3 `reaction()` options

Option	Default	Effect
`gap-h`	`1.5em`	Horizontal gap between items.
`gap-v`	`1.5em`	Vertical gap between rows.
`scale`	`1.0`	Uniform scale applied to the whole scheme.
`breakable`	`false`	Allow the scheme to split across pages.
`flow`	`"right"`	Writing direction: `"right"`, `"left"`, `"up"`, or `"down"` (left/up reflect the scheme).
`show-indices`	`false`	Default atom-index overlay for string SMILES molecules in this reaction.

18.4 `rxn-arrow()` options

Option	Default	Effect
`above`	`none`	Condition label (content or `mol()`) above a horizontal arrow, or right of a vertical arrow.
`below`	`none`	Condition label (content or `mol()`) below a horizontal arrow, or left of a vertical arrow.
`dir`	`auto`	Arrow direction: `"right"`, `"left"`, `"up"`, `"down"`, or `auto` (follows `reaction(flow:)`).
`kind`	`"single"`	Arrow style: `"single"`, `"equilibrium"`, `"equilibrium-filled"`, `"dashed"`, or `"wavy"`.
`scale`	`1.0`	Uniform arrow scale, including condition labels.
`color`	`auto`	Arrow color; `auto` inherits the surrounding text color.

18.5 Data helpers

Helper	Purpose

`mol-weight(smiles)`	Molecular weight in g/mol as a `float` errors on wildcards, abbreviations, and isotopes.
`smiles-inline(smiles, height:, baseline:, ..args)`	Molecule scaled and baseline-aligned for running text.
`smiles-cetz(smiles, name:, origin:, fg:, theme:, ..opts)`	Molecule as CeTZ elements with `atom-<i>` / `bond-<i>-<j>` / `center` anchors.

18.6 Mechanism helpers

Helper	Purpose
`mol(spec, label:, offset:, ..opts)`	Reaction item; `spec` as a string is rendered with addressable atoms; `offset` nudges in page coordinates; per-molecule `scale` also works in mechanism mode.
`cycle(..items, radius:, start:, clockwise:, scale:, reagent-bend:, arc-gap:)`	Catalytic cycle: species on a ring with arc arrows.
`step(label:, into:, out:, bend:, merge:, rotation:, *-offset:)`	A cycle arc: transformation label, reagent in/out, side-arrow bend, tangential merge, label rotation, and per-piece offsets.
`atom / bond / lp / species`	Atom-index references

	(optional `offset:`).
`arrow(from:, to:, label:, color:, bend:, angle:, half:, heads:, style:)`	Curly electron-pushing arrow; `heads` is `"end"` / `"both"` / `"none"`, `style` is `"solid"` / `"dashed"` / `"wavy"`.
`highlight(ref, fill:, stroke:, radius:, include-atoms:)`	Shade one atom/bond reference or an array of references.
`brackets(body, sup:, sub:)`	Square brackets with optional corner marks.

18.7 Label color names

Named styles for `{label|style}` — any of these plus any `#RRGGBB` hex:

Name	Swatch	Name	Swatch	Name	Swatch
red		orange		yellow	
brown		green		lime	
teal		cyan		blue	
navy		purple		pink	
black		gray		silver	
maroon		white			